NATIONAL FORENSIC SCIENCES UNIVERSITY, DELHI CAMPUS

P3: Vulnerability assessment and Penetration Testing

PRACTICAL 1: Network Reconnaissance and Scanning using Nmap

1. Aim

To perform network scanning and host reconnaissance using Nmap on Windows to demonstrate port scanning, operating system detection, UDP and TCP scanning, packet tracing, host discovery suppression, service version detection, and aggressive scanning.

2. Theory

Nmap (Network Mapper) is a free and open-source network scanning tool used for network discovery and security auditing. It helps identify active hosts, detect open and closed ports, identify running services, and estimate the operating system of a target machine.

Nmap is commonly used by network administrators and cybersecurity professionals to assess network security, troubleshoot connectivity problems, and identify potentially vulnerable services.

3. Requirements

- Windows computer
- Nmap installed
- Command Prompt or PowerShell
- Administrator privileges for certain scan types
- An authorized target, such as your own computer or a lab VM

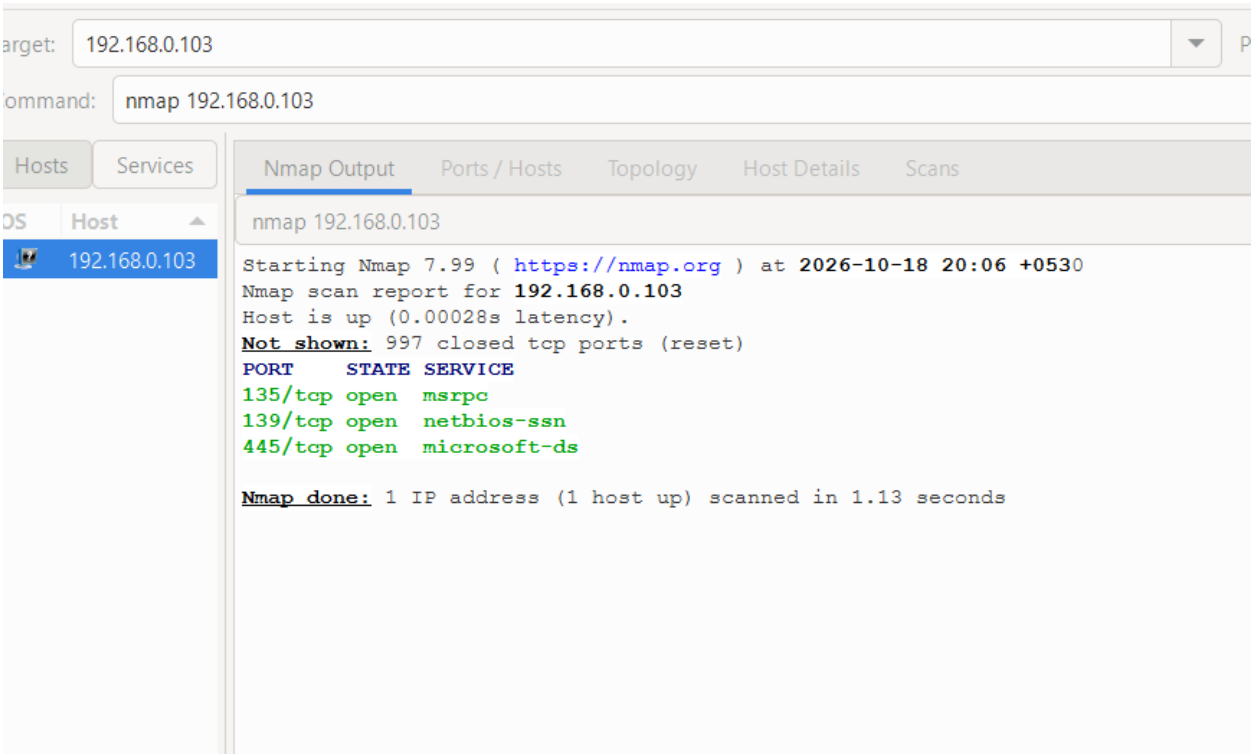
4. Procedure

Step 1: Basic Scan

Objective: Identify whether a host is reachable and check its common TCP ports.

1. Press `Windows + R`.
2. Type `cmd` and press Enter.
3. To find your own computer's IP address, type: ipconfig
4. Find the IPv4 Address under the network adapter you are using.
5. Run: nmap your ip address
6. Wait for the scan to finish and examine the open ports and host status.

Interpretation: The output lists the TCP ports Nmap checked and their states. A port marked open indicates that a service is listening on it.



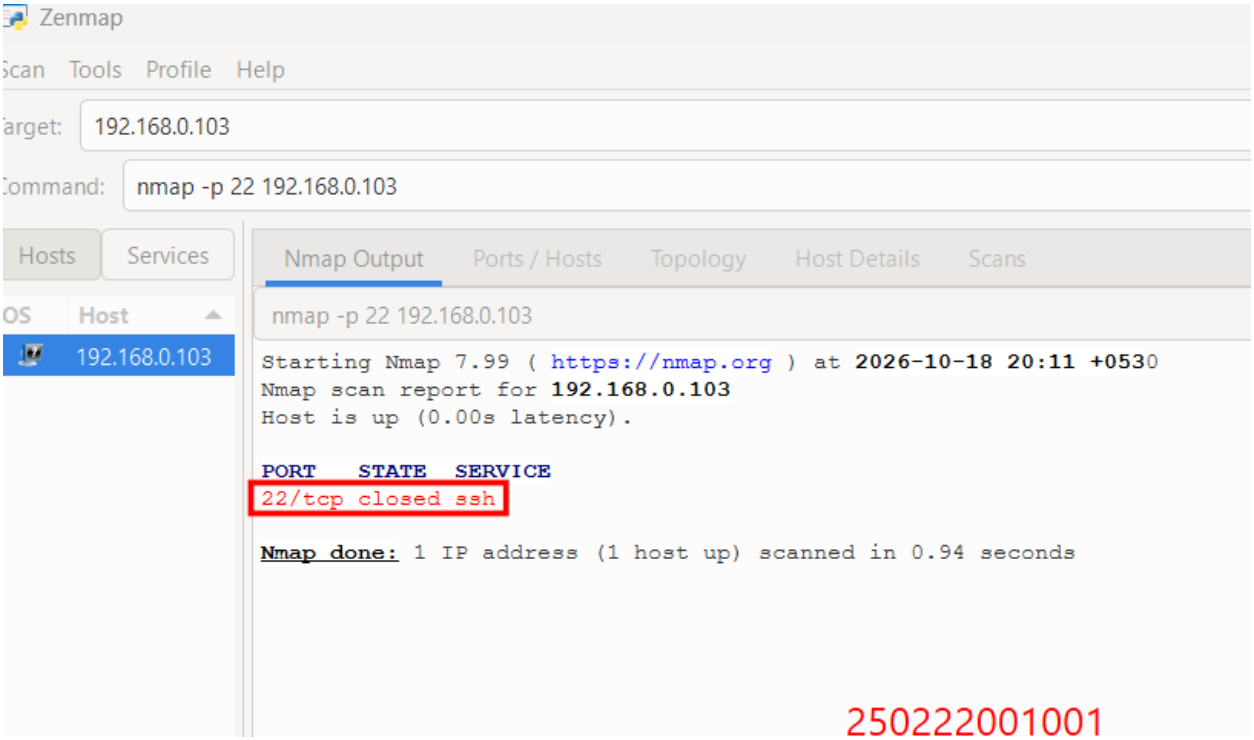
Step 2: Specific Port Scanning

Objective: Check the status of a particular port, such as port 22 (SSH).

1. Open Command Prompt.
2. Enter:

```
nmap -p 22 192.168.0.103
```

3. Press Enter.
4. Check the state of port 22 in the output.



Interpretation:

- Open: A service accepts connections on the port.
- `closed`: The host is reachable, but no service is listening on the port.
- `filtered`: Nmap cannot determine the port's state because filtering or a firewall interferes with the probe.

Step 3: Operating System Detection

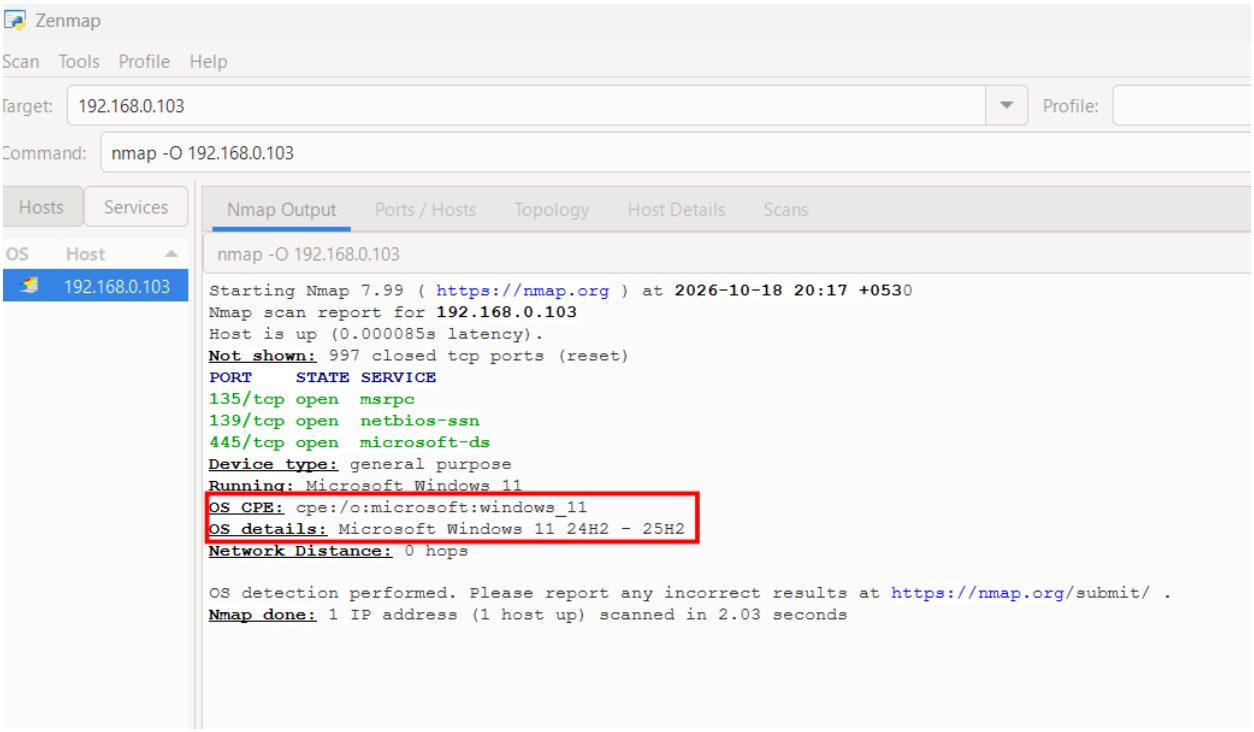
Objective: Attempt to identify the target operating system using Nmap's OS fingerprinting feature.

1. Open Command Prompt as Administrator.
2. Run:

```
nmap -O 192.168.0.103
```

3. Examine the output for OS details or OS detection warnings.

   Interpretation: Nmap may identify the operating system or provide possible OS guesses. Detection can fail or be inaccurate when the target does not provide enough information. Administrator privileges alone do not guarantee accurate detection.



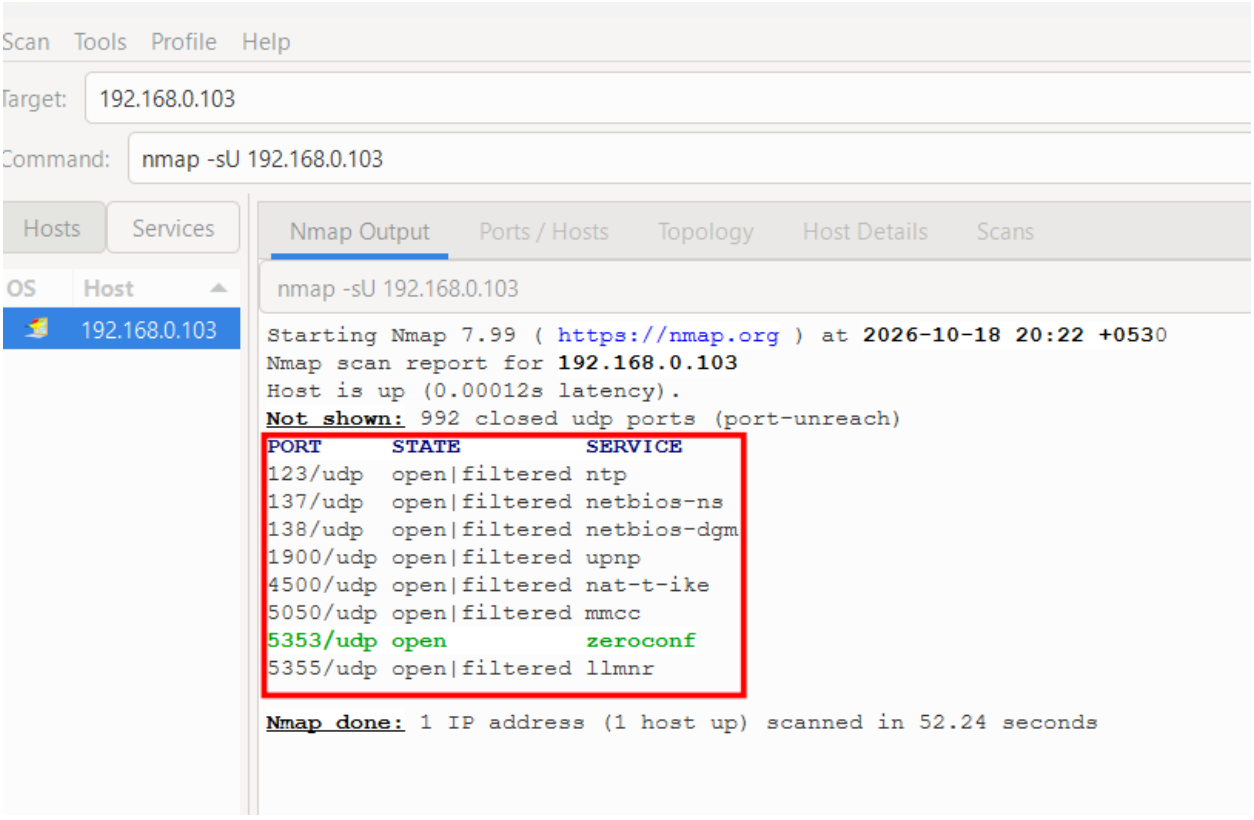
Step 4: UDP Scan

Objective: Check the state of UDP ports on the target.

1. Open Command Prompt as Administrator if required.
2. Run a limited UDP scan against your own computer:

`nmap -sU 192.168.0.103`

3. Wait for the scan to finish.
4. Note the states of the UDP ports.



Interpretation:

- Open: A UDP service responded.
- `closed`: A response indicates no service is listening.
- `open filtered`: Nmap cannot distinguish between an open port and filtering.

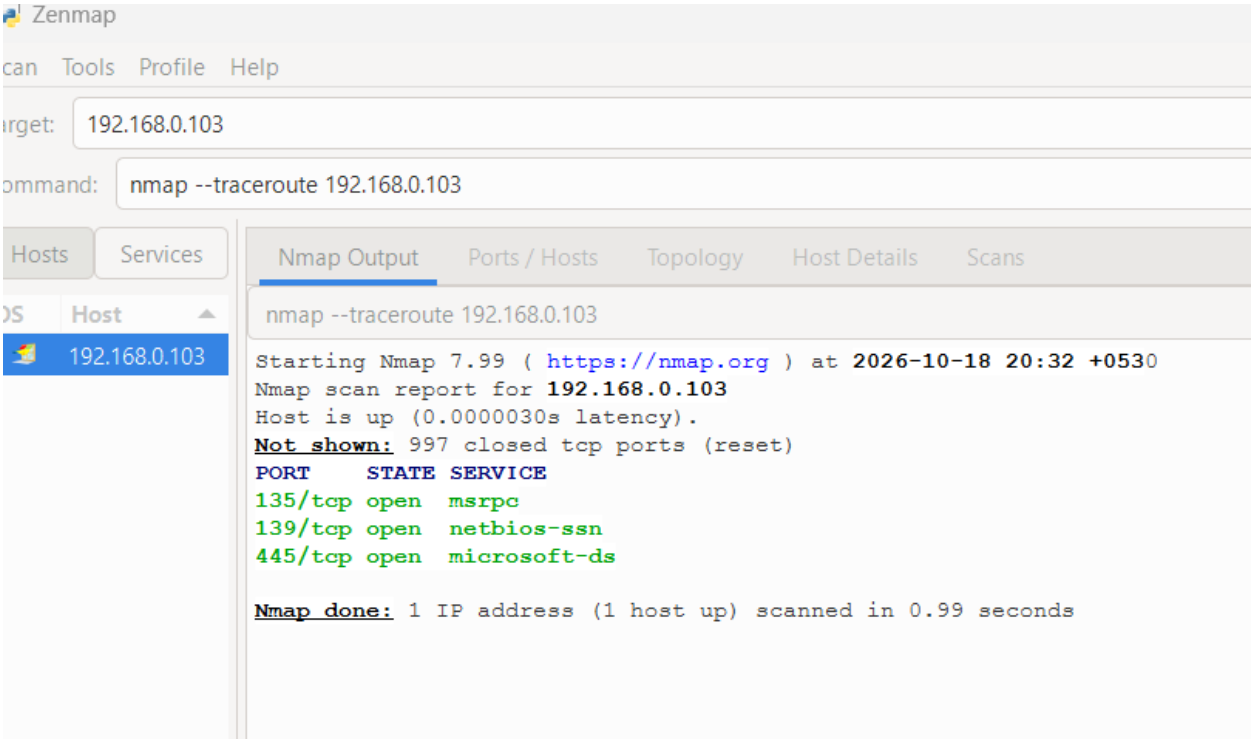
Step 5: Packet Tracing

Objective: Trace the network path to an authorized target.

1. Open Command Prompt.
2. Run:

```
nmap --traceroute 192.168.0.103
```

3. Examine the traceroute section.



Interpretation: The output may show hops and latency information. Because 192.168.0.103 refers to your own computer, it will not demonstrate a path through external routers. To observe multiple hops, use an authorized lab host on another network or a permitted destination.

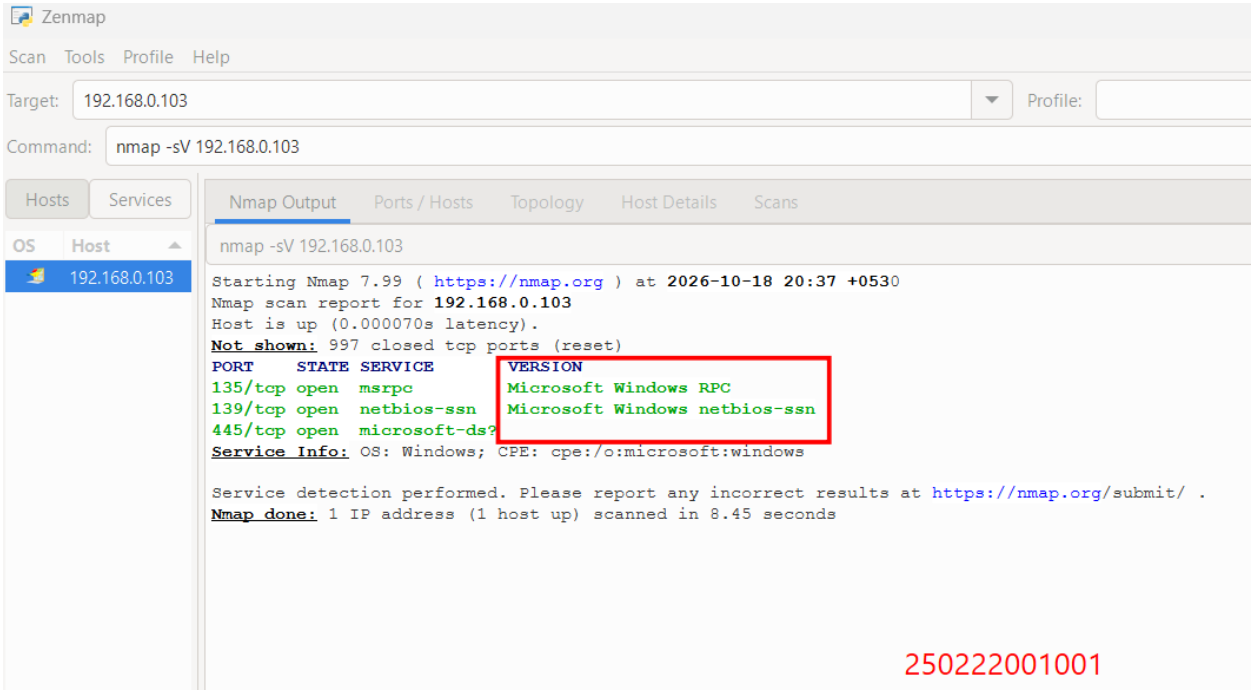
Step 6: Service Version Detection

Objective: Identify the services and possible software versions running on open ports.

1. Open Command Prompt.
2. Run:

```
nmap -sV 192.168.0.103
```

3. Review the port, state, service, and version columns.



Interpretation: Nmap probes detected open ports to identify the services running on them. Version information may be incomplete or unavailable if a service does not respond clearly.

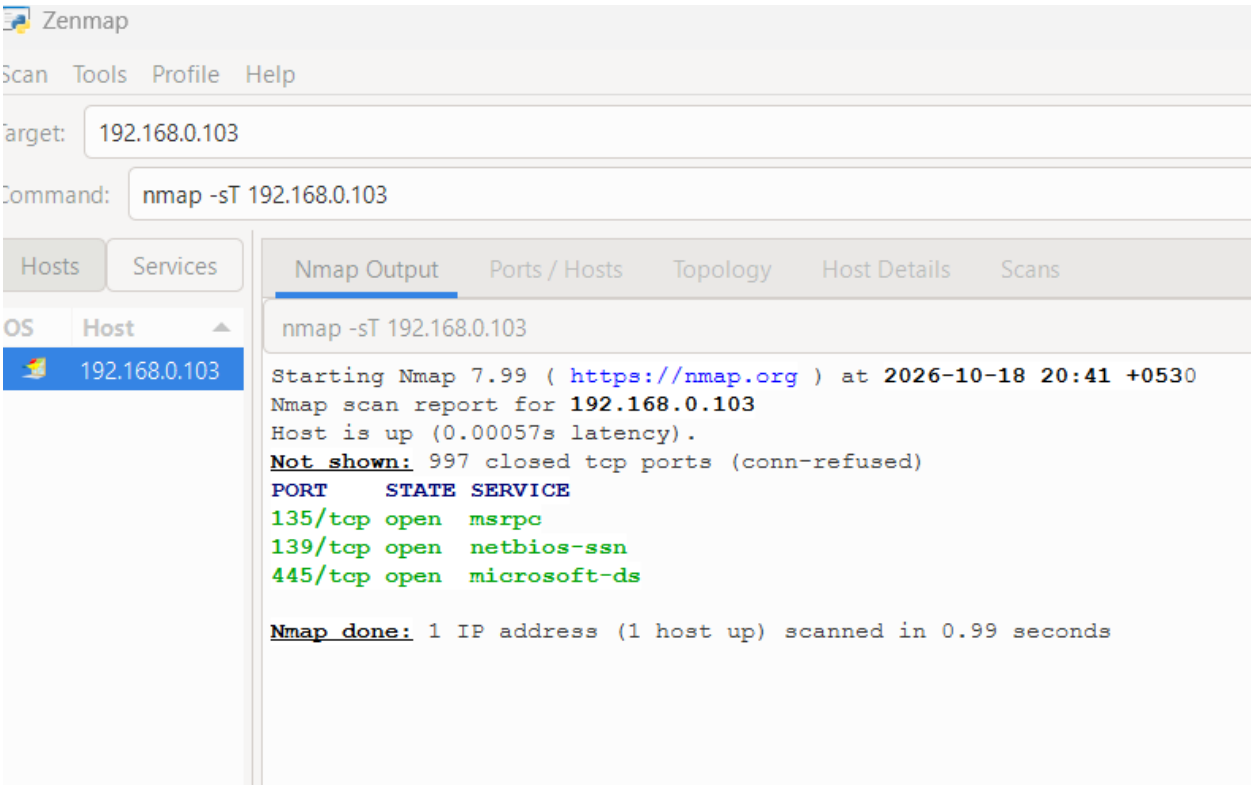
Step 7: TCP Connect Scan

Objective: Perform a TCP connect scan using the operating system's connection mechanism.

1. Open Command Prompt.
2. Enter:

```
nmap -sT 192.168.0.103
```

3. Press Enter.
4. Record the ports reported as open, closed, or filtered.



Interpretation: The `-sT` option attempts to establish full TCP connections to the target ports. It is useful when a SYN scan is unavailable or when demonstrating TCP connection behavior.

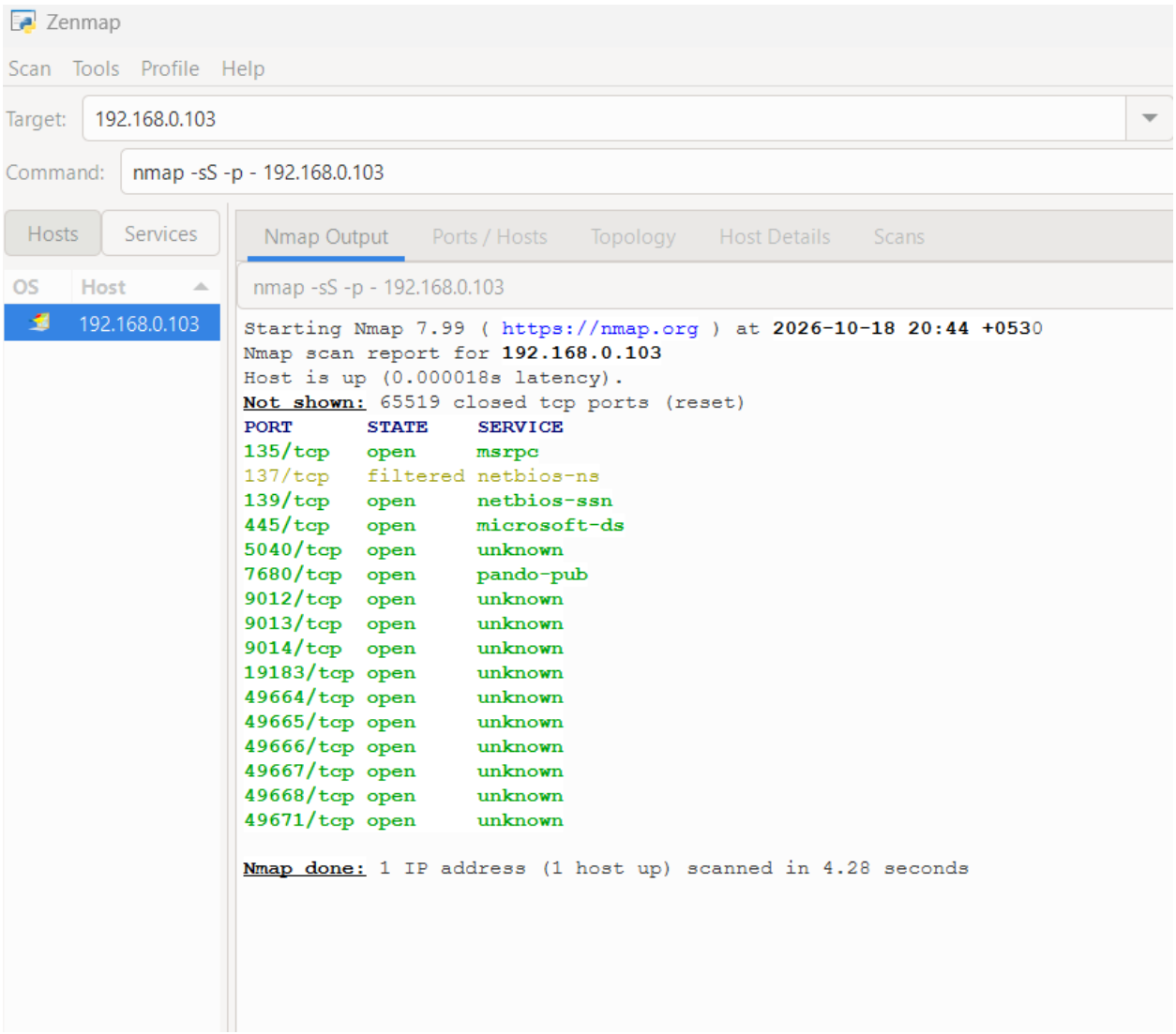
Step 8: TCP SYN Scan Across All Ports

Objective: Scan all 65,535 TCP ports on an authorized host.

1. Open Command Prompt as Administrator.
2. Run:

```
nmap -sS -p- 192.168.0.103
```

3. Wait for the scan to complete; scanning every port may take longer than a basic scan.
4. Record any open TCP ports.



Interpretation: The `-sS` option uses SYN probes, while `-p-` selects TCP ports 1 through 65535. The results help identify services that may not appear in a default scan.

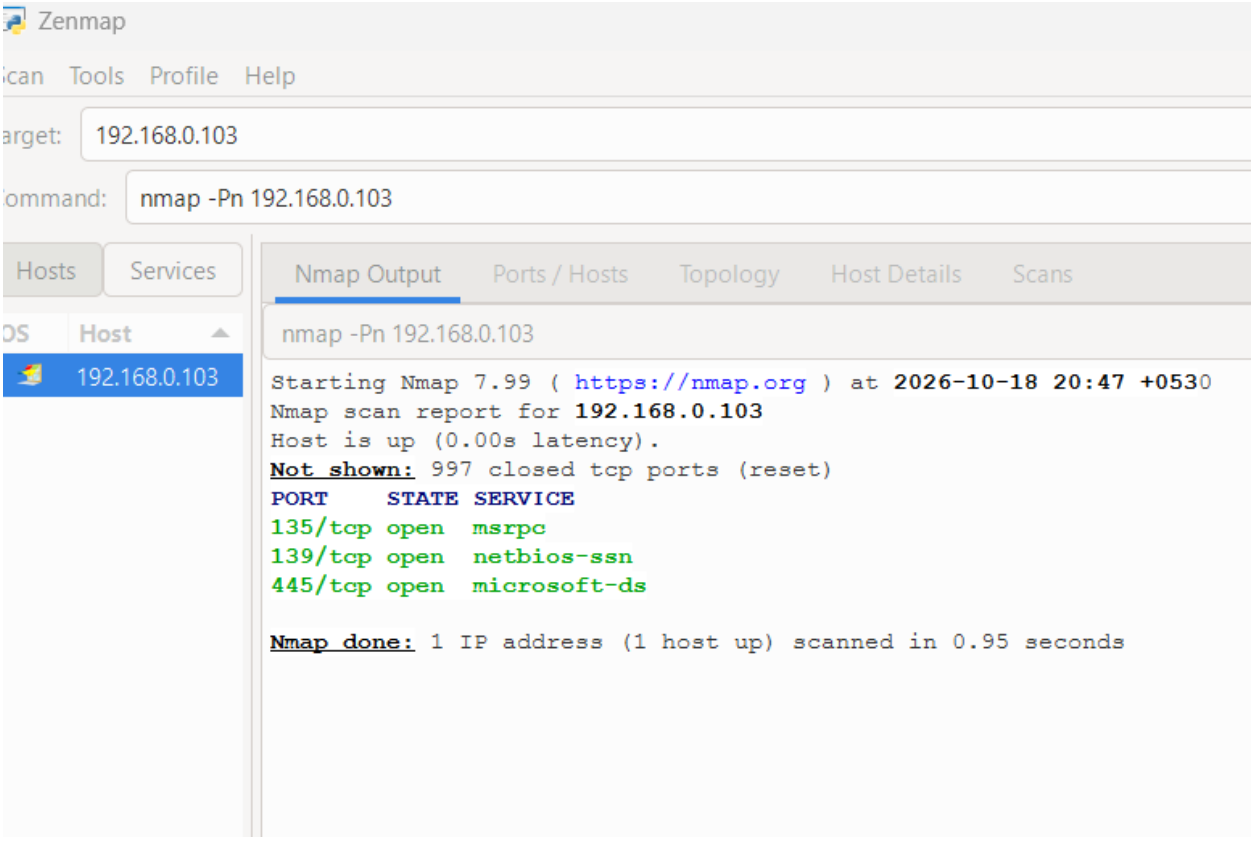
Step 9: Skip Host Discovery Using `-Pn`

Objective: Scan a host without relying on Nmap's usual preliminary host-discovery checks.

1. Open Command Prompt.
2. Run:

```
nmap -Pn 192.168.0.103
```

3. Review the port scan results.



Interpretation: The `-Pn` option tells Nmap to treat the target as online and proceed with scanning. It does not itself bypass a firewall or grant access to restricted ports.

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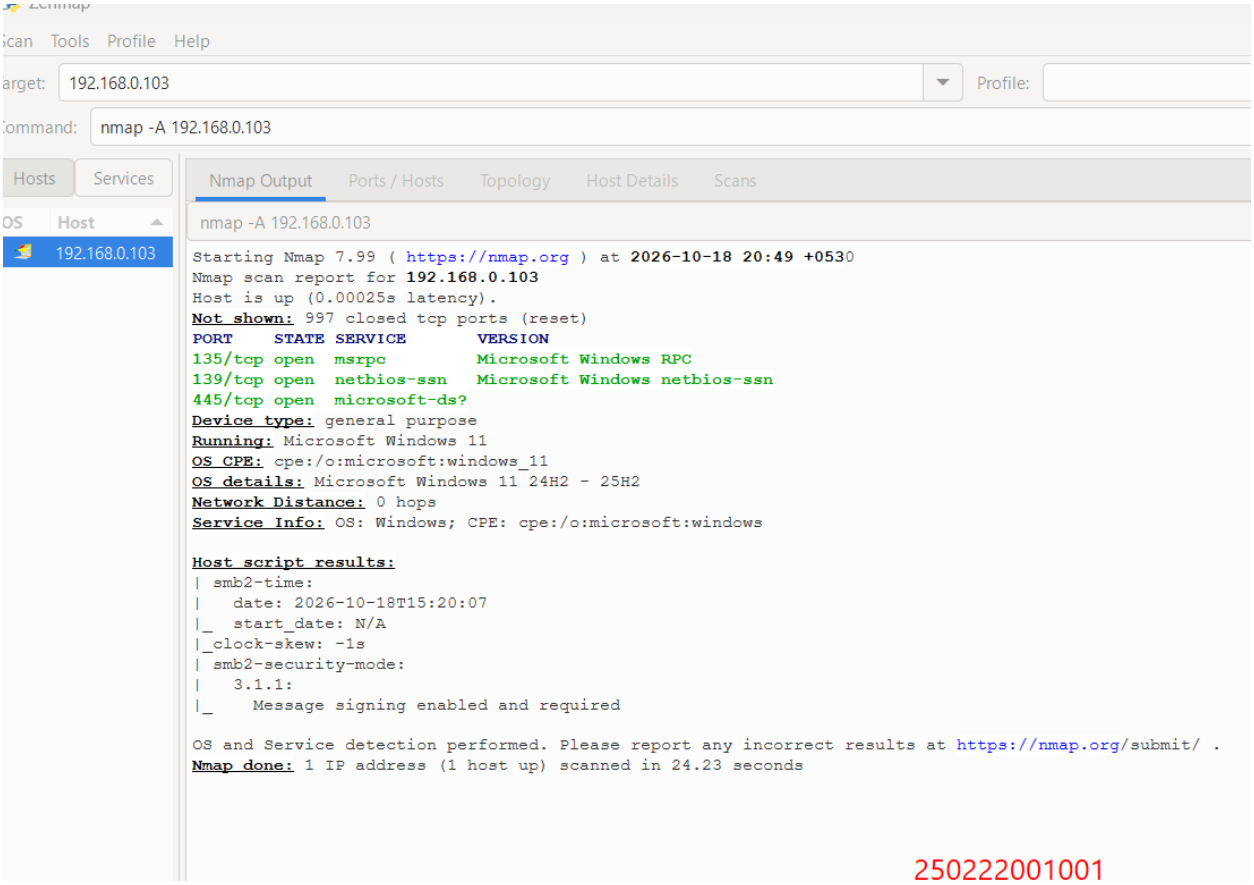
Step 10: Aggressive Scan

Objective: Combine several reconnaissance techniques into one scan.

1. Open Command Prompt as Administrator.
2. Run:

```
nmap -A 192.168.0.103
```

3. Wait for the scan to complete.
4. Review the output for OS guesses, service versions, script results, and traceroute information.



Interpretation: The `-A` option enables multiple detection features. Results depend on the target's responses and the privileges available. Use this option only on systems you are authorized to assess.

5. Observation

The following table shows what to record after performing each scan. Your actual output may differ from the sample PDF because you are scanning your own computer.

Scan	Command	Observation to record
Basic scan	`nmap 127.0.0.1`	Host status and common TCP port states
Specific port scan	`nmap -p 22 127.0.0.1`	State of TCP port 22
OS detection	`nmap -O 127.0.0.1`	OS identification or detection limitations
UDP scan	`nmap -sU -p 53,67,68,123,161 127.0.0.1`	UDP port states
Packet tracing	`nmap --traceroute 127.0.0.1`	Hops and latency, if reported
Service detection	`nmap -sV 127.0.0.1`	Detected services and versions
TCP connect scan	`nmap -sT 127.0.0.1`	Open, closed, and filtered TCP ports
All-port SYN scan	`nmap -sS -p- 127.0.0.1`	Open ports across the full TCP port range
Skip host discovery	`nmap -Pn 127.0.0.1`	Port scan results without normal host discovery
Aggressive scan	`nmap -A 127.0.0.1`	OS guesses, service details, script results, and traceroute

How to interpret common Nmap output

- `Host is up`: Nmap considers the target reachable.
- `PORT`: The port number and protocol being examined.
- `STATE`: Whether the port is open, closed, filtered, or in another reported state.
- `SERVICE`: The service commonly associated with the port or identified by Nmap.
- `VERSION`: The detected software or version, when available.
- `Not shown`: Ports omitted from the main output, often because they share the same state.

6. Result

Network reconnaissance and scanning techniques were studied and performed using Nmap on Windows. The experiment demonstrated basic host discovery, specific-port scanning, OS detection, UDP scanning, traceroute, service version detection, TCP connect scanning, full TCP SYN scanning, skipping host discovery, and aggressive scanning. The scan results can be used to understand network services and assess the security of authorized systems.

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